

DAVID PORATHUR

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SUMMARY

Final-year Electronics and Computer Science student with hands-on experience in building scalable cloud based applications, distributed systems across polyglot databases, LLM pipelines, and IoT infrastructure. Proficient in Python, C/C++, and Linux. Passionate about cybersecurity, systems design and building software that operates at scale.

EDUCATION

B.E. in Electronics and Computer Science (*Honors in Cybersecurity*) [2023 - 2027]

Fr. Conceicao Rodrigues College of Engineering, Bandra | CGPA: 8.2 of 10

- Elective Courses: Cryptography, Cloud Computing (AWS), System Security, Digital Forensics
- Certifications: Google Cloud [Cybersecurity Certificate](#), AWS Academy - [Cloud Architecting](#)

TECHNICAL SKILLS

- Linux, IoT Cloud
- SQL, Firestore, Qdrant, Neo4j
- C/C++, Embedded C
- Ollama, RestAPIs
- Python, FastAPI, Flask
- Docker, Github Actions
- AWS, Firebase, GCP
- VMWare ESXi on ARM Fling
- HTML, Tailwind CSS, Nextjs

PROFESSIONAL EXPERIENCE

Intern at National Critical Information Infrastructure Protection Centre - (NTRO) [JUNE 2025 - NOV 2025]

- Worked on multiple projects, architecting data ingestion pipelines integrating real-time OSINT feeds from 60+ sources, including OTX, Threatfox, MITRE, web crawlers, scrapers, etc.
- Developed the system infrastructure with a polyglot persistence, integrating Redis, PostgreSQL, Neo4j and Qdrant vector databases to ensure high availability and efficient querying of complex threat data.
- Fine-tuned specialized Large Language Models (LLMs), converting raw threat data into high-fidelity instruction sets for precise APT and entity extraction. Optimized performance by utilizing a Multi-GPU training setup and developing python scripts with multi-core processing for parallel data scraping.
- Delivered a complete, working prototype system featuring both a functional web dashboard for visual monitoring and a CLI for system control and real-time data analysis, used by the security and development teams.
- Collaborated with the security and development teams to improve the performance and reliability of the systems.

Intern at Oasis Luminaries - Product R&D Team [JAN 2025 - MAY 2025]

- Designed and developed a user-friendly web dashboard to control multiple IoT devices together, integrated APIs from various vendors as well as made 7+ RESTful APIs to establish links to the hardware system.
- Architected a custom IoT Gateway software for ESP32 and Raspberry Pi interfaced with an OpenWRT based IoT setup, successfully integrated multiple devices like advertising displays, smart lights, audio broadcast systems, etc.

PROJECTS

[CRCEats](#) | An App for the College Canteen

A full SaaS platform for managing college canteen ordering and operations end-to-end. Built with react native (expo), firebase and typescript. Integrated PhonePe payment gateway and QR based order verification. Implemented real-time order tracking, role-based access for users and canteen staff, and optimized performance with offline image caching.

[PanelSena](#) | A full-stack cloud-based digital signage platform (Open-sourced)

Designed for the centralized management of unlimited Raspberry Pi and Android displays. Leveraging a real-time Firebase infrastructure, the system incorporates secure device-based authentication, live playback synchronization, content scheduling, and comprehensive activity logging to ensure robust operational monitoring.

[Echo ATT&CK](#) | A Threat Intelligence Dashboard (Open-sourced)

A comprehensive cybersecurity threat intelligence platform that visualizes and analyzes MITRE ATT&CK framework data, providing security professionals with actionable insights into Advanced Persistent Threat (APT) groups and attack techniques. Implemented automated CI/CD pipelines using GitHub Actions to continuously sync and update threat data.

ACTIVITIES & ACHIEVEMENTS

- SIH Finalist 2024 | Winner at Hacksphere 2025 hackathon in cybersecurity domain | 2x Hackathon Winner
- College Communities: **Project Cell CRCE** [Technical Lead (2024-25) & Lead (2025-26)](3 Years)
Organised multiple sessions, workshops for the students. Built multiple innovative projects for the college. Designed and hosted a Capture-The-Flag (CTF) styled cybersecurity challenge. Led the organization of Prakalp 4.0 - A national level project exhibition competition, coordinating logistics and engagement for 115+ participating teams.